

Exploring the Interplay between Interaction Experience and Security Perception of Payment Authentication in Virtual Reality

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ABSTRACT

Users embrace the rapid development of virtual reality (VR) technology for everyday settings. These settings include payment, which makes user authentication necessary. Despite this need, there is a limited understanding of how users' unique experiences in VR contribute to their security perception. To understand this question, we designed probes of payment authentication, which are embedded in the routine payment of a VR game, to provoke participants' reactions from multiple angles.

Index Terms: Security and privacy—Human and societal aspects of security and privacy; Human-centered computing—Human computer interaction (HCI)—Empirical studies in HCI

1 INTRODUCTION

Virtual reality (VR) technology is seeing widespread adoption in everyday settings, including gaming and commerce. Payment features in VR empower these activities and contribute to the growth of VR economics [3]. To enable payments, VR systems access sensitive user data and assets, raising the need to authenticate users. VR introduces novel interaction experiences than traditional platforms (e.g., mobile phones) via its immersive virtual interface, where users interact with digital objects to perform routine activities in a virtual world, including payment. Recent research evaluated the usability of VR authentication [2] and has shown that the context in which authentication is used (e.g., where and for what purpose) affects how users perceive the security of authentication. For example, users feel insecure when using an ATM in a crowded space [6]. There is a critical need to understand **how the unique experience in VR contributes to users' security perception of payment authentication**. With such understanding, we can guide the future design of secure and usable authentication and better inform users of VR payment authentication's security properties. Our work aims to gain this understanding by investigating these related research questions.

- **RQ1:** What and how are the factors in users' interaction experiences influencing their security perceptions of payment authentication in VR?
- **RQ2:** What are the tensions in users' expectations for VR payment authentication in relation to **RQ1**?

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Figure 1: Our four probes and their usage context in VR.

We made a methodological contribution based on ideas of *technology probes* [4], where proof-of-concept interfaces uncover hidden phenomena in user interaction, to study payment authentication in VR. We designed four probes pertaining to authentication interactions – two variants of entering a PIN, tapping a virtual card, and signing a signature – to evaluate the user interaction experience and perceived security. In our study with 24 participants, we embedded these probes in a routine payment interaction for users when they play a VR archery game. Our initial findings highlight the need to support people's transition to VR payment and improve security awareness of VR payment authentication.

2 STUDY METHOD

2.1 Authentication Probes and Context

Design process. We iteratively developed and refined our design by considering the usage context and probes together. The final design is shown in Figure 1. While VR authentication methods are still in the early stage of commercialization, we identified three design dimensions of VR authentication: providing something you know, showing something you have, and proving who you are [7, 8]. They cover unique interaction patterns and inherent security properties. Our probes establish a mapping of (i.e., virtualize) other daily authentication interactions in the physical world into VR, which is a popular design choice in VR applications beyond payment. Note that our current work does *not* focus on comparing among these probes. Instead, these variations of probes helped us maximally capture different user interactions and identify common themes in the interplay between user experiences and security perceptions.

The four probes. We narrowed the design scope to the four probes: (1) floating PIN pad (PIN-F), (2) on-kiosk PIN pad (PIN-K), (3) tap-to-pay (TAP), and (4) signature (SIGN).

- **PIN-F** resembles the default virtual input interface of many VR platforms, where participants interact with a floating PIN pad that follows the participants' viewport. PIN-F also conceptualizes the idea of giving participants a personal and isolated virtual experience in the authentication.
- **PIN-K** presents a better mapping of physical world experiences than PIN-F by rendering its PIN pad on the kiosk where participants initiate and confirm payment. As such, participants do not experience a gap in the transition between authentication and other payment tasks, e.g., selecting items.
- **TAP** represents how people could own and use a personal virtual object as a unique token to authenticate. To pay, participants take out a virtual credit card from an inventory on their virtual body and tap it on the kiosk display. The kiosk checks whether the card is in proximity and being held by the owner.
- **SIGN** represents how participants draw a unique signature to give consent and prove their identity. It virtualizes the real-world signing processes. Participants grab a virtual pen from the kiosk and sign in the designated area; only after participants sign can they proceed to check out.

Our probes achieve these goals of the technology probe method [4] as follows. Their functionalities are *simple* – our designs only package the essential frontend interactions instead of the full back-end of a prototyped authentication system, including verifying an encrypted token. In this *early design phase*, we opted *not to optimize the usability* of probes in our implementation for provoking participants. This contrasts with a prototype, where usability is a primary aspect to optimize. We, however, made our design *flexible*, e.g., by allowing varying gestures, and support *logging* of participants' interactions, e.g., time consumed in authentication.

Study context: VR archery game We design a VR game – an archery contest – where participants trade in-game credits, using the probes to authenticate their payment. Participants earn credits by shooting virtual targets and consume credits when they refill arrows. To motivate the contest and make credit payment realistic, we match participants' credits with a small physical gift: the participant who scores the highest wins a "grand prize" (a 90 USD fitness tracker). We implemented our game using Unity (ver. 2021.3.4) and the VR Interaction Framework [1]. We ran the game on a commodity PC (GPU: Nvidia 3060Ti) connected to an Oculus Quest 2 headset. Participants interacted via the headset's hand-held controllers.

2.2 Study Design

Study procedure. Our within-subject study consists of two phases – enrollment and the main study. The purpose of enrollment is to familiarize participants with our setups. In the main study, participants interacted with our probes in the VR game. They finished four sessions; each included the game with an in-study survey. In each session, the participants paid using one of the probes (in a randomized order) to authenticate three times. The enrollment took about 20 minutes, and the main study was 1 hour, including breaks.

Instruments. Our instruments include (1) pre-study survey about participants' demographic backgrounds and technology affinity, (2) in-study survey about their general experience in the VR context and perceived usability of authentication probes, (3) game logs for recording participants' behaviors in the game, and (4) post-study survey to assess task engagement, security perception, and people's expectation for payment authentication.² Drawing from prior work, we study security perceptions around five aspects related to payment authentication, namely *consent*, *security*, *privacy*, *being*

²Our surveys are available at: https://osf.io/pt4mu/?view_only=57101bda15e84028ba7bbfeb151b5b55

alerted, and *in control*. We primarily conducted qualitative coding and analysis on participants' responses to open-ended questions.

Participants and recruitment. Our study was approved by the IRB-equivalent body of our organization. We recruited 24 participants. Each participant received compensation (a 30 USD gift). 18 out of 24 identified themselves as man (6 women). The average age of our participants is 29.6 years old (std: 4.8). 20 of them completed or were studying for a graduate degree. Most participants have a background in computer science. 20 participants had used VR before (mainly for gaming). But none of the 20 frequently used it.

3 FINDINGS

We observed that participants associated interaction experiences of authentication with better virtualization and more seamless, less disruptive interactions. These factors contribute to the security perception of VR authentication, similar to previous work on implicit authentication [5]. For instance, how authentication is virtualized in VR may bias participants when transferring prior understanding of authentication to VR, potentially leading to a false yet confident sense of security – some of them more confidently assumed that *"the technology behind tapping makes me [them] believe it is safe"* (P14). On the other hand, people expressed uncertainties regarding threats in both virtual and physical realms in the immersive VR, for instance, they might *"easily touch it [the virtual card] to an invisible system in the VR world"* (P19). The gamified context of VR also had the potential to alter participants' sensitivity to security risks of payment. Lastly, though people expected both security and usability, their expectations appear conflicting, e.g., the demand for a playful experience might diminish their sensitivity to security risks. The findings suggest future designs to improve security awareness of payment authentication, e.g., by optimizing security notices for VR.

4 CONCLUSIONS AND FUTURE WORK

We designed four probes of user authentication in the routine payment of a VR game to investigate how their interaction experiences affect users' security perception. This work motivates further efforts in different aspects to perform more in-depth qualitative analysis, quantitatively evaluate and compare authentication methods (including variables in design and implementation), explore the design space of authentication methods, and study how our findings can be generalized for other demographics and payment contexts.

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